

EBN 111/122
ELECTRICITY AND ELECTRONICS

Chapter 4
Circuit Theorems

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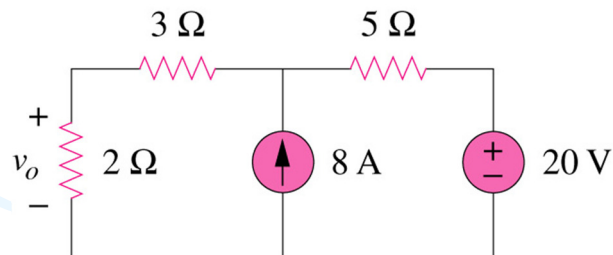
Circuit Theorems - Chapter 4

- 4.1 Motivation
- 4.2 Linearity Property
- 4.3 Superposition
- 4.4 Source Transformation
- 4.5 Thevenin's Theorem
- 4.6 Norton's Theorem
- 4.8 Maximum Power Transfer

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4.1 Motivation

If you are given the following circuit, are there any other alternative(s) to determine the voltage across 2Ω resistor?



What are they? And how?

Can you work it out by inspection?

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4.2 Linearity Property

It is the property of an element describing a linear relationship between cause and effect.

A linear circuit is one whose **output** is linearly related (or directly proportional) to its **input**.

Two properties of linearity

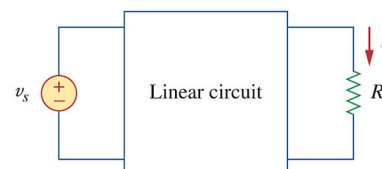
- Homogeneity (scaling)

$$v = iR \rightarrow kv = kiR$$

- Additivity

$$v_1 = i_1 R \text{ and } v_2 = i_2 R$$

$$\rightarrow v = (i_1 + i_2)R = v_1 + v_2$$

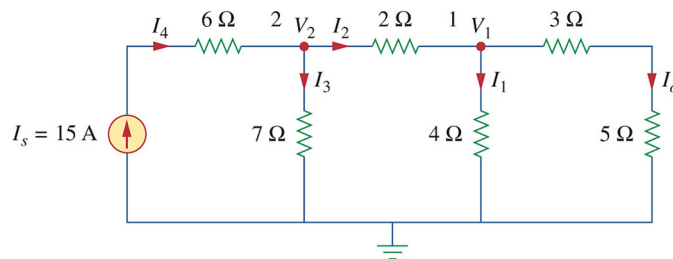


$$\begin{aligned} \text{Assume } v_s = 10\text{V} &\Rightarrow i = 2\text{ A} \\ \text{then: } v_s = 1\text{V} &\Rightarrow i = 0.2\text{ A} \\ v_s = 100\text{V} &\Rightarrow i = 20\text{ A} \end{aligned}$$

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Example 4.2

Assume $I_o = 1$ A and use linearity to find the actual value of I_o in the circuit.

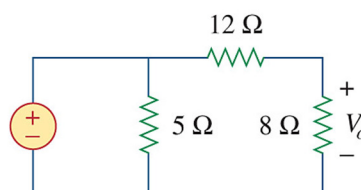


1. If $I_o = 1$ A: $V_1 = 1 \times (3 + 5) = 8$ V.
2. With $V_1 = 8$ V: $I_1 = 8/4 = 2$ A.
3. With $I_1 = 2$ A: $I_2 = I_o + I_1 = 1 + 2 = 3$ A (KCL)
4. With $I_2 = 3$ A: $V_2 = V_1 + I_2 \times 2 = 8 + 6 = 14$ V $I_2 = \frac{V_2 - V_1}{2}$
5. With $V_2 = 14$ V: $I_3 = 14/7 = 2$ A
6. With $I_3 = 2$ A: $I_4 = I_2 + I_3 = 3 + 2 = 5$ A (KCL)
7. Thus: $I_o/I_4 = I_o/I_s = 1/5$ A.
8. Due to linearity: $I_{o_actual} = 1/5 * I_{s_actual} = 3$ A

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PP 4.2

Assuming $V_o = 1$ V, use linearity to calculate the actual value of V_o .



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4.3 Superposition Theorem

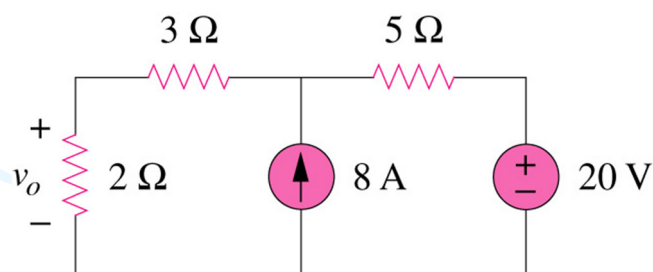
The voltage across (or current through) an element in a linear circuit is the algebraic sum of the voltage across (or currents through) that element due to EACH independent source acting alone.

The principle of superposition helps us to analyze a linear circuit with more than one independent source by calculating the contribution of each independent source separately.

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4.3 Superposition Theorem

We consider the effects of 8A and 20V one by one, then add the two effects together for final v_o .



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4.3 Superposition Theorem

STEPS

Steps to apply superposition principle

1. Turn off all independent sources except one source. Find the output (voltage or current) due to that active source using nodal or mesh analysis.
2. Repeat step 1 for each of the other independent sources.
3. Find the total contribution by adding algebraically all the contributions due to the independent sources.

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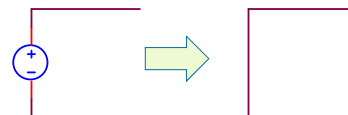
4.3 Superposition Theorem

Zeroing of Sources

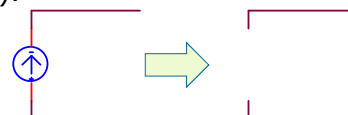
How do you turn off a source, i.e. make it zero?

1. Independent sources:

- Independent voltage sources are replaced by 0 V (short circuit ($R=0\ \Omega$)).



- Independent current sources are replaced by 0 A (open circuit ($R \rightarrow \infty\ \Omega$)).

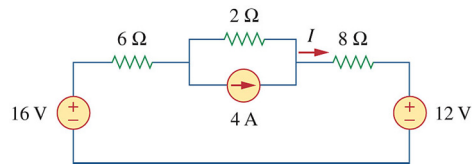


- ##### 2. Dependent sources are left intact because they are controlled by circuit variables.

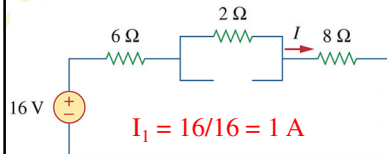
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PP 4.5:

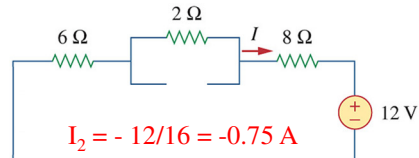
Use superposition to find I .



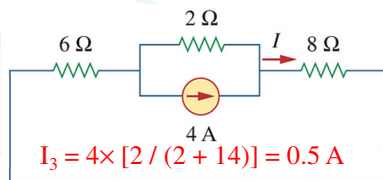
Switch off the 4 A and 12 V sources:



Switch off the 4 A and 16 V sources:



Switch off the 12 V and 16 V sources:

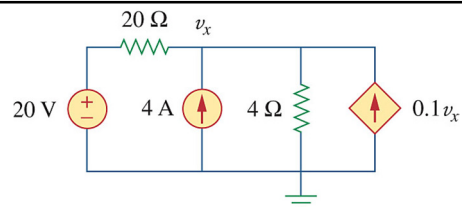


$$\begin{aligned} I &= I_1 + I_2 + I_3 \\ &= 1 + -0.75 + 0.5 \\ &= 0.75 \text{ A} \end{aligned}$$

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PP 4.4:

Use superposition to find v_x .



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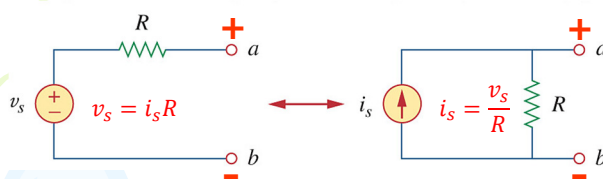
Ch. 4 L#2:

4.4 Source Transformations

- An equivalent circuit is one whose v - i characteristics are identical with the original circuit.
- It is the process of replacing a voltage source V_s in series with a resistor R by a current source I_s in parallel with a resistor R , or vice versa.

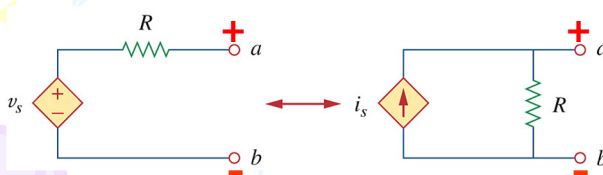
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4.4 Source Transformation



(a) Independent source transform

- The arrow of the current source is directed toward the positive terminal of the voltage source.



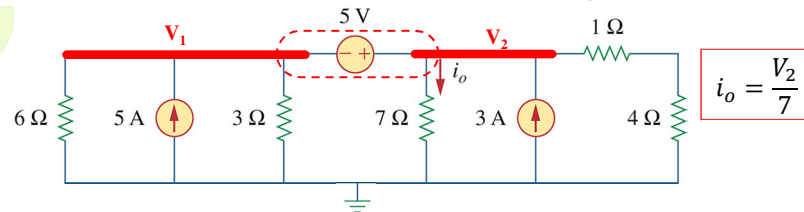
(b) Dependent source transform

- The source transformation is not possible when $R = 0$ for voltage source and $R = \infty$ for current source.

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PP 4.5b: Use nodal analysis and determine i_o .



Supernode: $V_2 - V_1 = 5 \quad \therefore V_1 = V_2 - 5 \quad (1)$

KCL @ Supernode:

$$\frac{V_1}{6} - 5 + \frac{V_1}{3} + \frac{V_2}{7} - 3 + \frac{V_2}{5} = 0 \quad (2)$$

$$\times 30 \quad 5(V_2 - 5) - 150 + 10(V_2 - 5) + \frac{30}{7}V_2 - 90 + 6V_2 = 0$$

$$5V_2 + 10V_2 + \frac{30}{7}V_2 + 6V_2 = +25 + 150 + 50 + 90$$

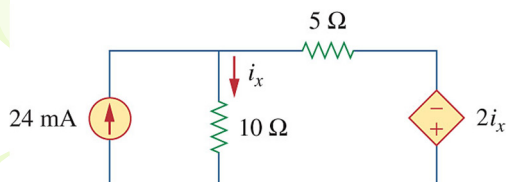
$$25.286V_2 = 315$$

$$\therefore V_2 = 12.457 \text{ V}$$

$$\therefore i_o = \frac{V_2}{7} = 1.78 \text{ A}$$

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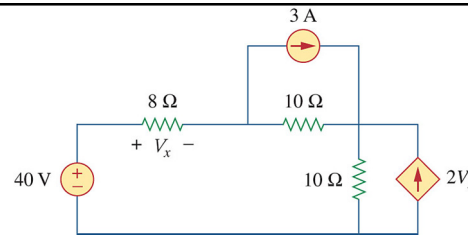
PP 4.7: Use source transformation and determine i_x .



Very IMPORTANT:
Do NOT transform the part
of the circuit with the
unknown variable!

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Problem 4.24: Determine v_x .



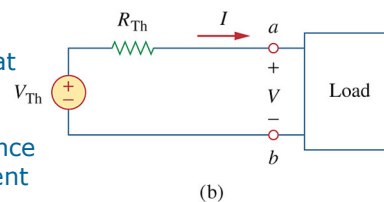
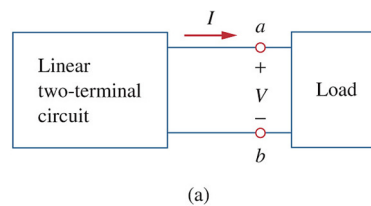
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Ch. 4 L#3: 4.5 Thevenin's Theorem

It states that a linear 2-terminal circuit (Fig. a) can be replaced by an equivalent circuit (Fig. b) consisting of a voltage source V_{TH} in series with a resistor R_{TH} .

where

- V_{TH} is the open-circuit voltage (V_{oc}) at the terminals.
- R_{TH} is the input or equivalent resistance at the terminals when the independent sources are turned off.

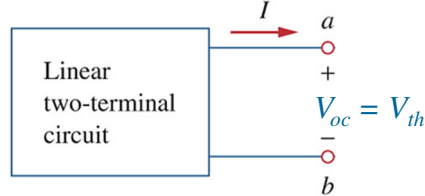


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4.5 Thevenin's Theorem

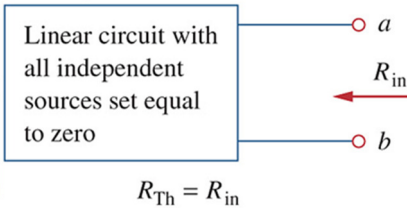
How to determine V_{th} and R_{th}

To determine V_{th} :



- V_{oc} = Open Circuit Voltage.
- Terminals a-b are terminals between which you want to determine the Thevenin equivalent.
- Can be anywhere in the circuit.

To determine R_{th} :



Method #1:

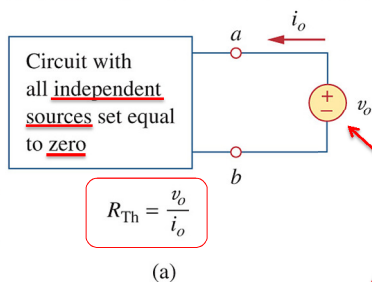
Zeroing of Sources:

If the circuit contains no dependent sources.

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4.5 Thevenin's Theorem

How to determine R_{th} Case 2

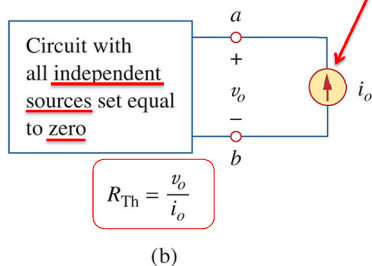


Method #2: Application of test sources, if the circuit does contain dependent sources.

Test Sources

MUCHO IMPORTANTE:

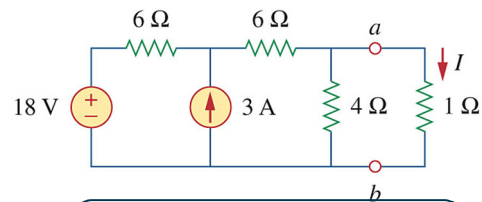
You cannot turn off dependent sources as they are controlled by circuit variables.



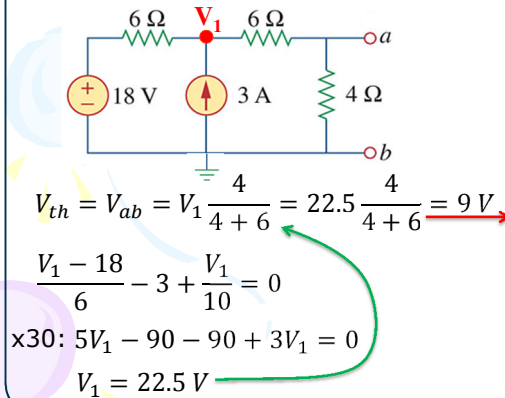
22

PP 4-8:

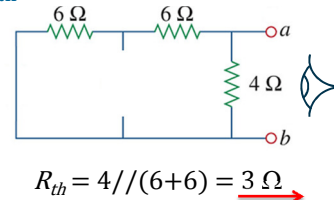
- a) Determine the Thevenin equivalent circuit to the left of terminals a-b.
- b) Determine I.



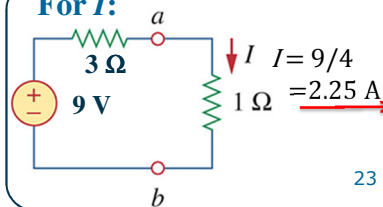
V_{th} : Calculate $V_{oc} = V_{ab}$:



R_{th} : Zero Ind. Sources:



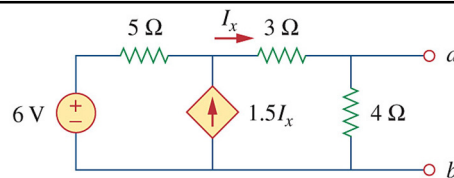
For I:



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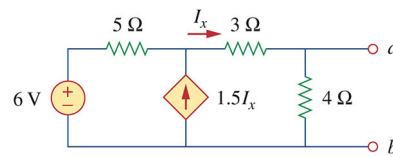
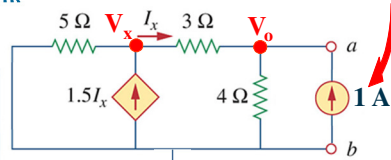
PP 4-9:

Determine the Thevenin equivalent circuit to the left of terminals a-b.



V_{th} :

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PP 4-9: continued **R_{th} : Method A** **R_{th} : Zero Ind. Sources + CURRENT Test Source: Need $V_{ab} = V_o$** 

$$\frac{V_x}{5} - 1.5I_x + \frac{V_x - V_o}{3} = 0$$

$$\text{But: } I_x = \frac{V_x - V_o}{3}$$

$$\frac{V_x}{5} - 1.5 \frac{V_x - V_o}{3} + \frac{V_x - V_o}{3} = 0$$

$$\times 15: 3V_x - 7.5(V_x - V_o) + 5(V_x - V_o) = 0$$

$$0.5V_x + 2.5V_o = 0 \quad (1)$$

$$\frac{V_o - V_x}{3} + \frac{V_o}{4} - 1 = 0$$

$$\times 12: 4(V_o - V_x) + 3V_o = 12$$

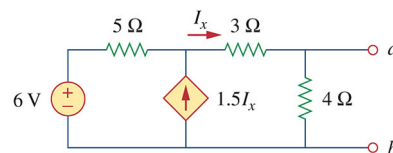
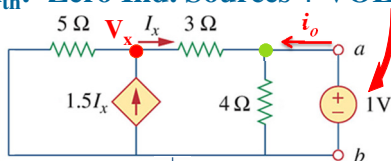
$$-4V_x + 7V_o = 12 \quad (2)$$

$$(2) + 8 \times (1): 7V_o + 20V_o = 12$$

$$V_o = 0.4 \text{ V}$$

$$\therefore R_{th} = \frac{V_o}{I_{test}} = \frac{V_o}{1} = 0.4 \Omega$$

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PP 4-9: continued **R_{th} : Method B** **R_{th} : Zero Ind. Sources + VOLTAGE Test Source: Need i_o** 

$$\frac{V_x}{5} - 1.5I_x + \frac{V_x - 1}{3} = 0$$

$$\text{But: } I_x = \frac{V_x - 1}{3}$$

$$\frac{V_x}{5} - 1.5 \frac{V_x - 1}{3} + \frac{V_x - 1}{3} = 0$$

$$\times 15: 3V_x - 7.5(V_x - 1) + 5(V_x - 1) = 0$$

$$0.5V_x + 2.5 = 0 \therefore V_x = -5 \text{ V}$$

$$\text{KCL: } \frac{V_x - 1}{3} + i_o - \frac{1}{4} = 0$$

$$\times 12: 4(-6) + 12i_o - 3 = 0$$

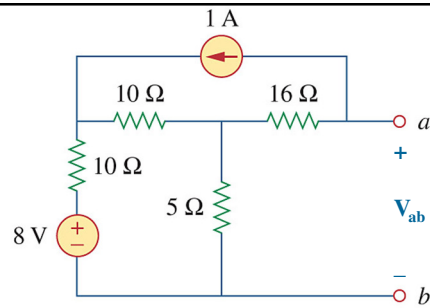
$$i_o = 2.25 \text{ A}$$

$$\therefore R_{th} = \frac{V_{test}}{i_o} = \frac{1}{2.25} = 0.4 \Omega$$

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Problem 4-39:

Determine the Thevenin equivalent circuit.



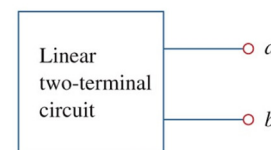
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Ch. 4 L#4: 4.6 Norton's Theorem

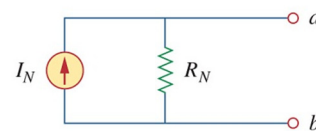
It states that a linear two-terminal circuit can be replaced by an equivalent circuit of a current source I_N in parallel with a resistor R_N .

Where

- I_N is the short circuit current through the terminals.
- R_N is the input or equivalent resistance at the terminals when the independent sources are turned off.



(a)



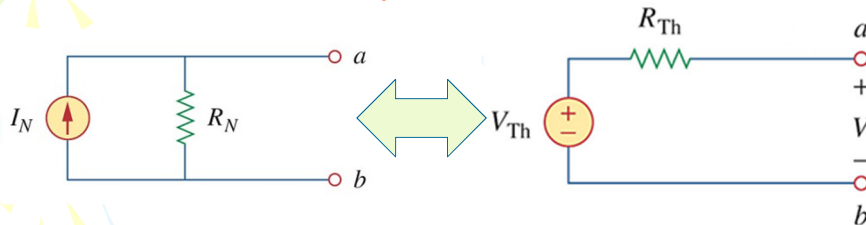
(b)

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4.6 Norton's Theorem

Norton vs Thevenin

Norton equivalent = Source transformation of the Thevenin equivalent and vice versa.



Where:

$$R_N = R_{th}$$

$$I_N = V_{th}/R_{th} = V_{th}/R_N$$

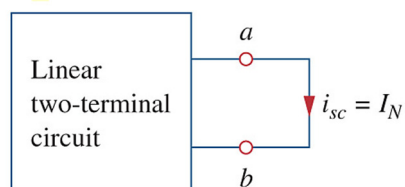
$$V_{th} = I_N R_{th} = I_N R_N$$

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4.6 Norton's Theorem

How to determine I_N and R_N

To determine I_N :



- i_{sc} = short circuit current.
- Terminals a-b are terminals between which you want to determine the Norton equivalent.
- Can be anywhere in the circuit.

To determine R_N :

Methods #1 and #2 as with R_{th} since $R_N = R_{th}$.

Method #3:

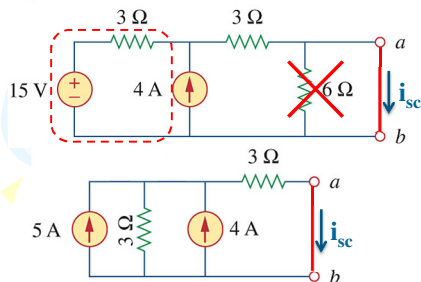
- Not in this textbook, but you will find it in other.
- Short circuit terminals a-b & do not zero any sources.
- $R_N = R_{th} = V_{oc} / i_{sc}$
- Works for circuits with or without dependent sources.

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PP 4-11:

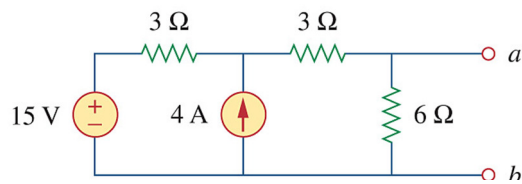
Determine the Norton equivalent circuit to the left of terminals a-b.

I_N : Calculate $i_{sc} = I_N$:



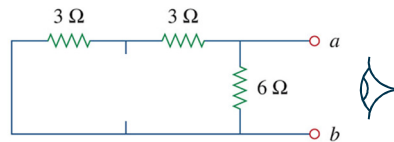
Hence 9 A splitting between two 3 Ω resistors:

$$I_N = 9/2 = 4.5 \text{ A}$$



R_N : Method #1

Only independent sources: Zero them.



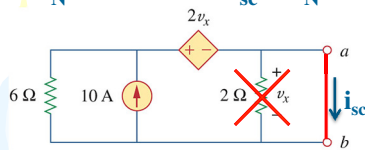
$$R_N = 6/(3+3) = 3 \Omega$$

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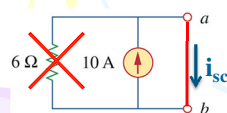
PP 4-12:

Determine the Norton equivalent circuit.

I_N : Calculate $i_{sc} = I_N$:

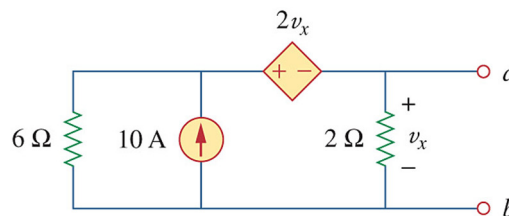


Observe: SC makes $v_x = 0 \text{ V}$ and hence also $2v_x = 0$.



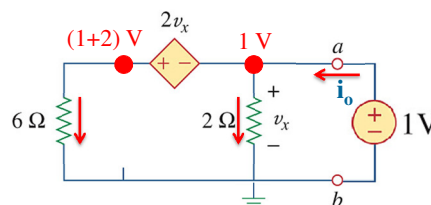
6 Ω shorted, hence no current through the 6 Ω :

$$i_{sc} = 10 \text{ A}$$



R_N : Method #2:

Zero Ind. Sources + Test Source



KCL @ 1 V node:

$$i_o = i_2 + i_6$$

$$i_o = \frac{1}{2} + \frac{1+2}{6} = \frac{1}{2} + \frac{1}{2} = 1 \text{ A}$$

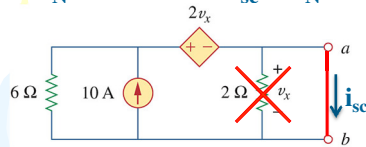
$$R_N = \frac{V_{test}}{i_o} = \frac{1}{1} = 1 \Omega$$

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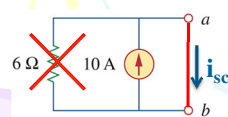
PP 4-12:

Determine the Norton equivalent circuit.

I_N : Calculate $i_{sc} = I_N$:

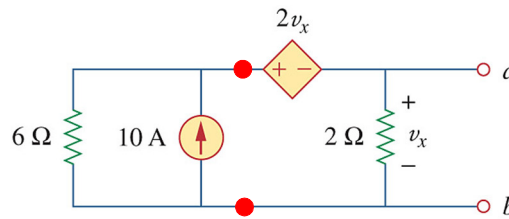


Observe: SC makes $v_x = 0$ V and hence also $2v_x = 0$.



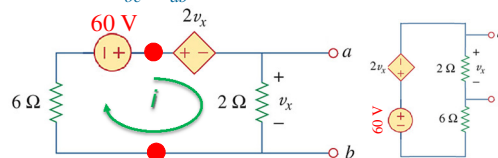
6 Ω shorted, hence no current through the 6 Ω :

$$i_{sc} = 10 \text{ A}$$



R_N : Method #3: $R_{th} = V_{oc}/i_{sc}$

Find $V_{oc} = V_{ab}$



Into current flowing, all elements are now in series, use Voltage Div to get v_x :

$$v_x = \frac{2}{2+6} (60 - 2v_x)$$

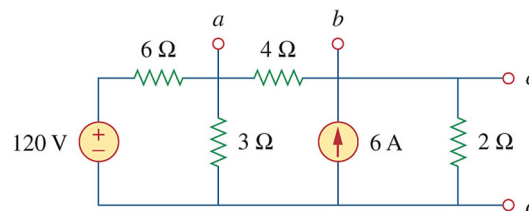
$$v_x = v_{oc} = V_{th} = 10 \text{ V}$$

$$R_N = R_{th} = \frac{v_{oc}}{i_{sc}} = 1 \Omega$$

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Problem 4-51:

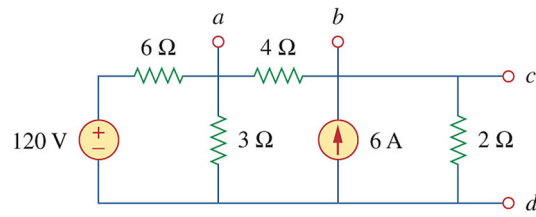
Determine the Norton equivalent circuit between terminals a-b.



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Problem 4-51 (continued):

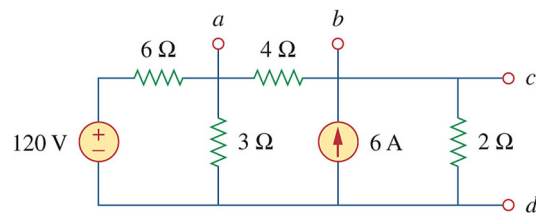
Determine the Norton equivalent circuit between terminals a-b.



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Problem 4-51 (continued):

Determine the Norton equivalent circuit between terminals a-b.



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Ch. 4 L#5:

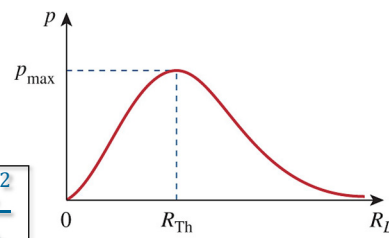
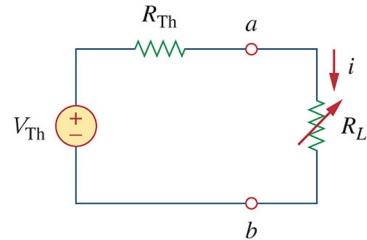
4.8 Maximum Power Transfer

If the entire circuit is replaced by its **Thevenin equivalent** except for the load, the power delivered to the load is:

$$P = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L$$

For maximum power dissipated in R_L , P_{max} , for a given R_{Th} and V_{Th} :

$$R_L = R_{th} \rightarrow P_{max} = \frac{V_{th}^2}{4R_{th}} = \frac{V_{RL}^2}{R_{th}}$$



The power transfer profile with different R_L

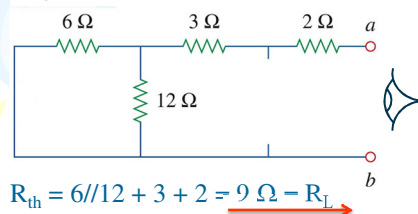
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Ex 4-13:

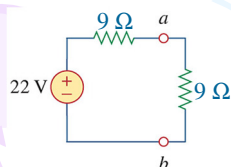
Determine R_L for maximum power transfer as well as what P_{max} will be.

Method: Determine the Thevenin circuit between a-b. I.e. V_{th} and R_{th} .

R_{th} : Zeroing of sources:

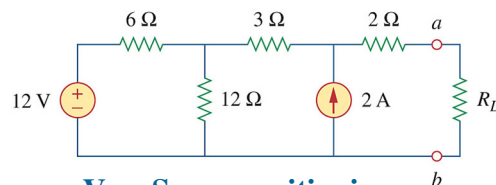


$$R_{th} = 6//12 + 3 + 2 = 9\Omega - R_L$$

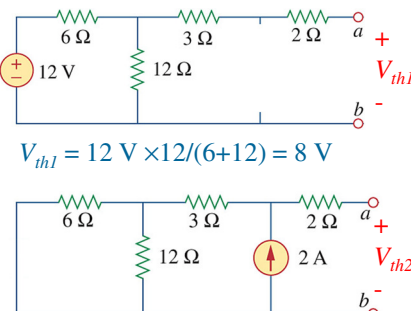


$$P_{max} = \frac{22^2}{4 \times 9} = 13.44\text{ W}$$

V_{th} : Super positioning:



$$V_{th1} = 12\text{ V} \times 12/(6+12) = 8\text{ V}$$



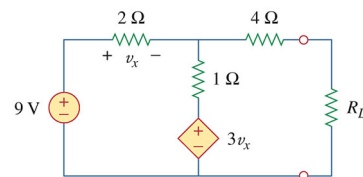
$$V_{th2} = 2\text{ A} \times ((6//12) + 3) = 2 \times 7 = 14\text{ V}$$

$$\therefore V_{th} = 8 + 14 = 22\text{ V}$$

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PP 4-13:

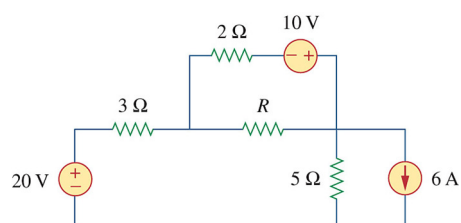
Determine R_L for maximum power transfer as well as what $P_{\max}$ will be.



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Problem 4.66:

Determine the maximum power that can be delivered to resistor R .



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